

Power over Ethernet Standards

IEEE 802.3bt (Type 4)

IEEE 802.3bt Type 4 delivers up to 90 watts of DC power over all four pairs of a balanced twisted-pair cable. It requires a minimum of Cat 5e cabling and is backwards compatible with 802.3af and 802.3at powered devices.

- Maximum power: 90 W
- Pairs used: all four
- Minimum cable: Cat 5e

IEEE 802.3at (Type 2)

IEEE 802.3at Type 2 (PoE+) delivers up to 30 W. It requires two pairs and is classified as Midspan or Endpoint PSE. The maximum DC resistance per pair set is 12.5 ohms.

Cable Temperature

Bundles of 192 or more PoE cables require derating of the conductor current capacity. Temperature rise at the bundle centre shall not exceed 10 degrees Celsius above ambient.